



Wandering Black Holes

Some Black holes aren't at their galaxy's center, but instead appear to roam the outskirts. <https://digit.in/june19-41>



Hole in the Milky Way

Something enormous appears to have torn a hole in part of the Milky Way's halo. <https://digit.in/june19-43>

IMAGING A BLACK HOLE

How to photograph an object from which even light cannot escape?

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lack holes are incredibly dense regions, where matter has been squeezed together into

incredibly small spaces. There are various kinds of black holes, with the smallest being about the size of a single atom. The mass of an entire mountain can be squeezed into this tiny radius, and such black holes were created at the birth of the universe. Bigger black holes can be created by the collapse of stars, and are called stellar black holes. A stellar black hole can have a mass that is up to 20 times the mass of the Sun. The largest black holes are known as supermassive black holes, and are typically found in the hearts of galaxies. Such black holes can be more massive than 4 million Suns, and can easily fit a few million Earths within their radius. When two black holes merge, they create an even more massive black hole. The LIGO instrument observed gravitational waves from such a black hole merger back in 2015. Black holes are regions of space from which no form of electromagnetic waves can escape,



Credit: STScI/AURA

including light. Around the black hole is a region known as the event horizon, and anything that passes beyond it travels forever towards the center of the black hole. Stephen Hawking had theoretically shown that a certain kind of emission was possible from the event horizon, and this is known as Hawking Radiation. As there is no light escaping from the black hole, it is difficult to capture it using equipment that works on the basis of capturing gases. However, the intense gravitational field around

the black holes churns gas and plasma. This gas releases energy, which can be observed using astronomical instruments.

THE EVENT HORIZON TELESCOPE

The closest known stellar black hole to Earth is V616, which is about 3,000 light years away. This is a black hole of relatively low mass, between 9-13 times the mass of the sun. The closest supermassive black hole to the Earth is at the centre of the Milky